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# Diophantine Equations

*An Original Olympiad Practice Set*

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**Original content.** These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad problems on Diophantine equations, ordered easiest to hardest. They span linear coin-style equations, difference-of-squares factoring, the unit-fraction divisor count, infinite descent and modular obstructions, Mordell-type cubics, Pell-flavoured equations with solution orbits, and an exponential Diophantine finale. Each rewards a structural insight rather than computation, and every numerical answer has been independently checked with sympy. Every closed-form answer was checked symbolically/numerically. Free to share for study.

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# Problems

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**1.** DIFFICULTY 2/5 · LINEAR DIOPHANTINE, BEZOUT / GCD, BOUNDING NONNEGATIVE SOLUTIONS

A vending machine accepts only 7-cent and 11-cent tokens. Determine all ways to insert a nonnegative number of each kind of token so that the total value is exactly 100 cents; that is, find every solution in nonnegative integers of

$$7x + 11y = 100.$$

**2.** DIFFICULTY 2/5 · DIFFERENCE OF SQUARES, FACTORING, PARITY OBSTRUCTION, DIVISOR PAIRING

Find the number of solutions in positive integers  $(x, y)$  of

$$x^2 - y^2 = 2024,$$

and list them all.

**3.** DIFFICULTY 3/5 · UNIT FRACTIONS, SIMON'S FAVORITE FACTORING TRICK, DIVISOR COUNTING, NUMBER OF DIVISORS OF  $N^2$

Let  $n$  be a positive integer. Prove that the number of ordered pairs of positive integers  $(x, y)$  satisfying

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}$$

equals  $d(n^2)$ , the number of positive divisors of  $n^2$ . As a check, verify the formula for  $n = 6$ .

**4.** DIFFICULTY 3/5 · INFINITE DESCENT, MODULAR ARITHMETIC MOD 3, QUADRATIC FORMS

Prove that the only integer solution of

$$x^2 + y^2 = 3z^2$$

is  $x = y = z = 0$ .

**5.** DIFFICULTY 4/5 · MORDELL EQUATION, MODULAR OBSTRUCTION MOD 4, FACTORING  $Y^2+1$ , PRIMES 3 MOD 4 AND SUMS OF TWO SQUARES

Prove that the equation

$$y^2 = x^3 + 7$$

has no solutions in integers.

**6.** DIFFICULTY 4/5 · PELL-LIKE EQUATION, NORM FORM  $X^2 - 2Y^2$ , FUNDAMENTAL UNIT, SOLUTION ORBITS / CLASSES

Find all solutions in nonnegative integers  $(x, y)$  of the Pell-like equation

$$x^2 - 2y^2 = 7,$$

describing the complete solution set explicitly.

**7.** DIFFICULTY 3/5 · COMPLETING THE SQUARE, FACTORING OVER INTEGERS, BOUNDED FACTOR PAIRS, CONSECUTIVE INTEGERS TRAP

Find all integer pairs  $(x, y)$  with

$$x^2 = y^2 + y + 1.$$

**8.** DIFFICULTY 5/5 · EXPONENTIAL DIOPHANTINE, MODULAR OBSTRUCTION MOD 4, FACTORING  $3^X - 1$ , CASE ANALYSIS ON PARITY

Find all pairs of nonnegative integers  $(x, y)$  satisfying

$$3^x - 2^y = 1.$$

## Solutions

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**1. Answer: The unique solution is  $(x, y) = (8, 4)$ .**

Since  $\gcd(7, 11) = 1$  divides 100, integer solutions exist. Reduce modulo 7:  $11y \equiv 100 \pmod{7}$  becomes  $4y \equiv 2 \pmod{7}$ . Multiplying by the inverse of 4 modulo 7 (which is 2, since  $4 \cdot 2 = 8 \equiv 1$ ) gives  $y \equiv 4 \pmod{7}$ . So  $y \in \{4, 11, 18, \dots\}$ . For  $y \geq 0$  we also need  $x = \frac{100-11y}{7} \geq 0$ , i.e.  $11y \leq 100$ , so  $y \leq 9$ . The only value of  $y$  with  $y \equiv 4 \pmod{7}$  and  $0 \leq y \leq 9$  is  $y = 4$ , giving  $x = \frac{100-44}{7} = 8$ . Hence  $(x, y) = (8, 4)$  is the only nonnegative solution. ■

**Second method.** General theory: a particular solution is  $(x_0, y_0) = (8, 4)$ , and the full integer family is  $x = 8 + 11t$ ,  $y = 4 - 7t$  for  $t \in \mathbb{Z}$ . Requiring  $x \geq 0$  gives  $t \geq -\frac{8}{11}$ , i.e.  $t \geq 0$ ; requiring  $y \geq 0$  gives  $t \leq \frac{4}{7}$ , i.e.  $t \leq 0$ . The two bounds force  $t = 0$ , recovering  $(8, 4)$  uniquely.

**2. Answer: There are exactly 4 solutions:**  $(45, 1)$ ,  $(57, 35)$ ,  $(255, 251)$ ,  $(507, 505)$ .

Factor the left side:  $x^2 - y^2 = (x - y)(x + y) = 2024$ . Write  $x - y = a$  and  $x + y = b$  with  $0 < a < b$  and  $ab = 2024$  (both factors are positive since  $x > y > 0$ , and  $a \neq 0$ ). Recovering  $x = \frac{a+b}{2}$ ,  $y = \frac{b-a}{2}$  as integers requires  $a$  and  $b$  to have the same parity. They cannot both be odd, because then  $ab$  would be odd while 2024 is even. Hence both must be even. (This is consistent:  $2024 = 2^3 \cdot 11 \cdot 23$  is divisible by 4, so a product of two even numbers can equal it.) Write  $a = 2a'$ ,  $b = 2b'$ ; then  $4a'b' = 2024$ , so  $a'b' = 506 = 2 \cdot 11 \cdot 23$ . The divisor pairs  $(a', b')$  with  $a' < b'$

**Second method.** Counting shortcut: a difference of two squares equals  $N$  iff  $N \not\equiv 2 \pmod{4}$ . When solutions exist, the number of positive solutions equals the number of factorizations  $N = ab$  with  $a < b$  and  $a, b$  of the same parity.

**3. Answer: Proof**

Clearing denominators:  $n(x + y) = xy$ , i.e.  $xy - nx - ny = 0$ . Add  $n^2$  to both sides to factor (Simon's Favorite Factoring Trick):

$$(x - n)(y - n) = n^2.$$

For  $x, y$  to be positive we must have both factors positive: from  $\frac{1}{x} = \frac{1}{n} - \frac{1}{y} < \frac{1}{n}$  we get  $x > n$ , so  $x - n > 0$ , and likewise  $y - n > 0$ . Set  $u = x - n$ ,  $v = y - n$ ; then  $u, v$  are positive integers with  $uv = n^2$ . Each ordered factorization  $uv = n^2$  gives a unique ordered pair  $(x, y) = (u + n, v + n)$  of positive solutions, and conversely. The number of ordered factorizations of  $n^2$  into two positive factors is exactly the number of choices of  $u \mid n^2$ , namely  $d(n^2)$ . Hence the count is  $d(n^2)$ . Check  $n = 6$ :  $n^2 = 36 = 2^2 \cdot 3^2$ , so  $d(36) = (2 + 1)(2 + 1) = 9$ . The nine solutions correspond to the nine divisors of 36; for instance  $u = 1 \Rightarrow (7, 42)$ ,  $u = 6 \Rightarrow (12, 12)$ ,  $u = 36 \Rightarrow (42, 7)$ , and so on. ■

**Second method.** Symmetry remark: writing  $n^2 = \prod p_i^{2a_i}$ , each exponent in  $n^2$  is even, so  $d(n^2) = \prod (2a_i + 1)$  is always odd. This matches the structure: solutions pair up as  $(x, y) \leftrightarrow (y, x)$  except for the single symmetric solution  $x = y = 2n$  (from  $u = v = n$ ), so the total must be odd — consistent with  $d(n^2)$  being odd.

**4. Answer: Proof**

Suppose a nonzero integer solution exists. Among all solutions with  $(x, y, z) \neq (0, 0, 0)$ , choose one minimizing  $z^2$  (well-defined since values are nonnegative integers; note  $z \neq 0$ , else  $x^2 + y^2 = 0 \Rightarrow x = y = 0$ ). Work modulo 3. Squares modulo 3 are 0 or 1. Since  $3z^2 \equiv 0 \pmod{3}$ , we need  $x^2 + y^2 \equiv 0 \pmod{3}$ . The only way two values from  $\{0, 1\}$  sum to 0 (mod 3) is  $0 + 0$ ; thus  $x \equiv 0$  and  $y \equiv 0 \pmod{3}$ . Write  $x = 3x_1$ ,  $y = 3y_1$ . Substituting:  $9x_1^2 + 9y_1^2 = 3z^2$ , so  $3(x_1^2 + y_1^2) = z^2$ . Then  $3 \mid z^2$ , hence  $3 \mid z$ ; write  $z = 3z_1$ . Substituting again:  $3(x_1^2 + y_1^2) = 9z_1^2$ , i.e.  $x_1^2 + y_1^2 = 3z_1^2$ . So  $(x_1, y_1, z_1)$  is again a solution, with  $z_1^2 = z^2/9 < z^2$  and still  $z_1 \neq 0$  (since  $z \neq 0$ ) — a strictly smaller nontrivial solution. This contradicts minimality. Therefore no nonzero solution exists, and  $(0, 0, 0)$  is the only one. ■

**Second method.** Descent via the 3-adic valuation, no minimality needed: suppose  $(x, y, z)$  is a nonzero solution and let  $k = v_3(z)$  be the exact power of 3 dividing  $z$ . The argument above shows every solution forces  $3 \mid x$ ,  $3 \mid y$ ,  $3 \mid z$  and produces a solution  $(x/3, y/3, z/3)$ . Iterating,  $3^m$  divides  $z$  for every  $m \geq 0$ , which is impossible for a nonzero integer  $z$ . Hence  $z = 0$ , and then  $x = y = 0$ .

## 5. Answer: Proof

Suppose integers  $x, y$  satisfy  $y^2 = x^3 + 7$ . We first show  $x$  is odd. Work modulo 4: squares are 0 or 1 (mod 4). If  $x$  were even, then  $x^3$  is divisible by 8, so  $x^3 \equiv 0 \pmod{4}$  and  $y^2 = x^3 + 7 \equiv 0 + 7 \equiv 3 \pmod{4}$ , which is impossible. Hence  $x$  is odd. Now add 1 to both sides:

$$y^2 + 1 = x^3 + 8 = (x + 2)(x^2 - 2x + 4).$$

Consider the second factor  $x^2 - 2x + 4 = (x - 1)^2 + 3$ . Since  $x$  is odd,  $x - 1$  is even, so  $(x - 1)^2 \equiv 0 \pmod{4}$ , giving  $x^2 - 2x + 4 \equiv 3 \pmod{4}$ . This factor is a positive integer  $\equiv 3 \pmod{4}$  (indeed  $(x - 1)^2 + 3 \geq 3$ ), so it must have at least one prime factor  $p \equiv 3 \pmod{4}$  (a product of primes that are all  $\equiv 1 \pmod{4}$ , together with possible factors of 2, can never be  $\equiv 3 \pmod{4}$ ; and an odd number has no factor 2). That prime  $p$  divides  $y^2 + 1$ , so  $y^2 \equiv -1 \pmod{p}$ . But  $-1$  is a quadratic residue modulo an odd prime  $p$  only when  $p \equiv 1 \pmod{4}$ ; for  $p \equiv 3 \pmod{4}$  the congruence  $y^2 \equiv -1$  has no solution. Contradiction. Therefore the equation has no integer solutions. ■

**Second method.** One can phrase the key step via sums of two squares directly: by Fermat's two-squares criterion, no prime  $p \equiv 3 \pmod{4}$  divides  $a^2 + 1$  for any integer  $a$  (such a  $p$  must appear to an even power in any sum of two squares, yet it would divide  $a^2 + 1$  exactly once is already contradicted by  $a^2 \equiv -1$  being insoluble mod  $p$ ). Since the factor  $x^2 - 2x + 4 \equiv 3 \pmod{4}$  forces a prime divisor  $p \equiv 3 \pmod{4}$  of  $y^2 + 1$ , we again reach a contradiction.

**6. Answer: All nonnegative solutions are generated from the two base solutions (3, 1) and (5, 3) by repeatedly applying  $(x, y) \mapsto (3x + 4y, 2x + 3y)$ . Explicitly the solutions are the two interleaving sequences (3, 1), (13, 9), (75, 53), ... and (5, 3), (27, 19), (157, 111), ...**

Solutions of  $x^2 - 2y^2 = 7$  live in the ring  $\mathbb{Z}[\sqrt{2}]$  as elements  $x + y\sqrt{2}$  of norm  $N(x + y\sqrt{2}) = x^2 - 2y^2 = 7$ . The associated Pell equation  $u^2 - 2v^2 = 1$  has fundamental solution  $(u, v) = (3, 2)$ , i.e. the fundamental unit  $\varepsilon = 3 + 2\sqrt{2}$  of norm 1. Multiplying any solution  $\alpha$  of norm 7 by  $\varepsilon$  preserves the norm:  $N(\alpha\varepsilon) = N(\alpha)N(\varepsilon) = 7 \cdot 1 = 7$ . Concretely,  $(x + y\sqrt{2})(3 + 2\sqrt{2}) = (3x + 4y) + (2x + 3y)\sqrt{2}$ , giving the map  $T : (x, y) \mapsto (3x + 4y, 2x + 3y)$ . To find the base solutions, one bounds the smallest representative in each class; checking directly, the smallest nonnegative solutions are (3, 1) and (5, 3) (indeed  $9 - 2 = 7$  and  $25 - 18 = 7$ ), and these are inequivalent under  $T$  (the map sends  $(3, 1) \rightarrow (13, 9)$ , skipping  $(5, 3)$ ). Every nonnegative solution is obtained from one of these two by applying  $T$  finitely many times. Applying  $T$ : from (3, 1) we get (13, 9), (75, 53), (437, 309), ...; from (5, 3) we get (27, 19), (157, 111), ... Together these two orbits comprise all nonnegative solutions. ■

**Second method.** Recurrence form: since both  $x$  and  $y$  coordinates satisfy the linear recurrence  $a_{k+1} = 6a_k - a_{k-1}$  (the trace of  $\varepsilon^2 = 17 + 12\sqrt{2}$  is 6), one can present the answer as: the  $x$ -values in each class obey  $x_{k+1} = 6x_k - x_{k-1}$  and similarly for  $y$ . For the class of (3, 1):  $x : 3, 13, 75, 437, \dots$ ; for the class of (5, 3):  $x : 5, 27, 157, \dots$ . This avoids quoting  $\mathbb{Z}[\sqrt{2}]$  and is checkable by direct substitution into  $x^2 - 2y^2 = 7$ .

**7. Answer: The solutions are  $(x, y) \in \{(1, 0), (-1, 0), (1, -1), (-1, -1)\}$ . In particular  $|x| = 1$  always, and  $y \in \{0, -1\}$ .**

Multiply by 4 to complete the square on the right:

$$4x^2 = 4y^2 + 4y + 4 = (2y + 1)^2 + 3.$$

Rearrange into a difference of squares:

$$(2x)^2 - (2y + 1)^2 = 3,$$

i.e.

$$(2x - 2y - 1)(2x + 2y + 1) = 3.$$

Both factors are integers whose product is 3, so the unordered factor pair  $\{2x - 2y - 1, 2x + 2y + 1\}$  is one of  $\{1, 3\}$  or  $\{-1, -3\}$ . Note the two factors sum to  $4x$ . For  $\{1, 3\}$ :  $\text{sum} = 4 \Rightarrow x = 1$ ; the difference  $(2x + 2y + 1) - (2x - 2y - 1) = 4y + 2$  equals  $\pm 2$ , giving  $y = 0$  or  $y = -1$ . For  $\{-1, -3\}$ :  $\text{sum} = -4 \Rightarrow x = -1$ ; similarly  $4y + 2 = \pm 2$  gives  $y = 0$  or  $y = -1$ . Collecting:  $(x, y) \in \{(1, 0), (1, -1), (-1, 0), (-1, -1)\}$ . ■

**Second method.** Squeeze argument: for  $y \geq 1$  we have  $y^2 < y^2 + y + 1 < (y + 1)^2 = y^2 + 2y + 1$ , so  $x^2$  lies strictly between two consecutive squares and cannot be a perfect square. For  $y \leq -2$ , set  $y = -m$  with  $m \geq 2$ : then  $y^2 + y + 1 = m^2 - m + 1$ , and  $(m - 1)^2 = m^2 - 2m + 1 < m^2 - m + 1 < m^2$ , again strictly between consecutive squares. So the only candidates are  $y \in \{-1, 0\}$ , both giving  $x^2 = 1$ , hence  $x = \pm 1$ . This yields the same four solutions.

**8. Answer: The only solutions are  $(x, y) = (1, 1)$  and  $(2, 3)$ .**

We seek  $3^x = 2^y + 1$  with  $x, y \geq 0$ . **Small cases.** If  $y = 0$ :  $3^x = 2$ , no. If  $y = 1$ :  $3^x = 3 \Rightarrow x = 1$ ; solution  $(1, 1)$ . If  $y = 2$ :  $3^x = 5$ , no. If  $y = 3$ :  $3^x = 9 \Rightarrow x = 2$ ; solution  $(2, 3)$ . We now show no solution has  $y \geq 4$  (in fact we handle all  $y \geq 2$  uniformly). **Step 1 (mod 4, parity of x).** For  $y \geq 2$ ,  $2^y \equiv 0 \pmod{4}$ , so  $3^x \equiv 1 \pmod{4}$ . Since  $3 \equiv -1 \pmod{4}$ ,  $3^x \equiv (-1)^x$ , forcing  $x$  even. Write  $x = 2k$  with  $k \geq 1$ . **Step 2 (factor).** Then  $2^y = 3^{2k} - 1 = (3^k - 1)(3^k + 1)$ . The two factors  $3^k - 1$  and  $3^k + 1$  differ by 2 and are both even; their gcd divides 2, so one of them is  $\equiv 2 \pmod{4}$  (exactly divisible by 2) and the other carries the remaining factors of 2. Since the product  $2^y$  has only the prime factor 2, each factor must itself be a power of 2. Two powers of 2 differing by 2 are only 2 and 4. Hence  $\{3^k - 1, 3^k + 1\} = \{2, 4\}$ , giving  $3^k = 3$ , so  $k = 1$  and  $x = 2$ . Then  $2^y = 3^2 - 1 = 8$ , so  $y = 3$ . **Conclusion.** The case  $y \geq 2$  yields only  $(2, 3)$ , and the cases  $y \in \{0, 1\}$  yield only  $(1, 1)$ . Therefore the complete solution set is  $\{(1, 1), (2, 3)\}$ . ■

**Second method.** Alternative finish after Step 1 via lifting the exponent (LTE): for  $y \geq 2$  write  $x = 2k$ . Applying LTE to  $3^{2k} - 1$  at the prime 2:  $v_2(3^{2k} - 1) = v_2(3^2 - 1) + v_2(3) + v_2(2k) - 1 = 1 + 2 + v_2(2k) - 1 = 2 + v_2(2k)$ . But the entire value  $3^{2k} - 1 = 2^y$  is a pure power of 2, so it must equal  $2^{2+v_2(2k)}$  with no odd factor. For  $k \geq 2$ , the odd part of  $3^{2k} - 1$  exceeds 1 (e.g.  $k = 2$  gives  $80 = 2^4 \cdot 5$ , with odd factor 5), so the equality fails. Only  $k = 1$  survives, giving  $(x, y) = (2, 3)$ , consistent with the first method.